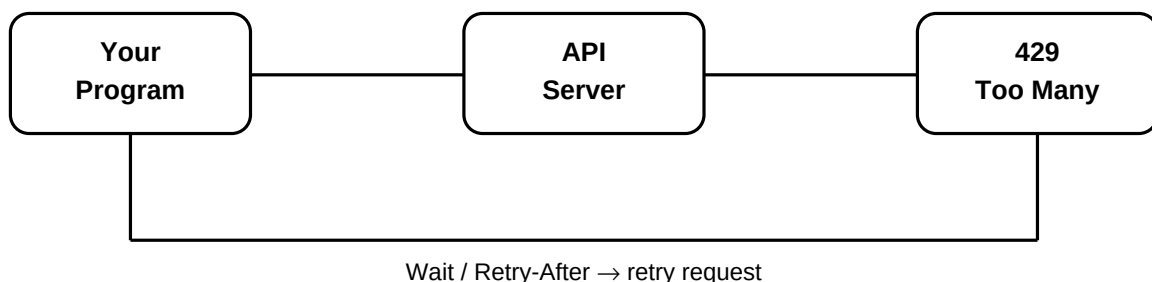


MINI ASSIGNMENT – 2

Name	Dilli Guna Sekhar
Register No.	99230041049
Branch	CSE (AIML)
Date	11-09-2026

1. If an API returns 429, what should your program do?

Answer: A 429 status code means **Too Many Requests**. The program should not retry immediately. It should wait before trying again. If the server provides a **Retry-After** value, the program should follow it. A limited retry strategy with increasing delays (exponential backoff) can also be used. This prevents repeated requests from making the problem worse.

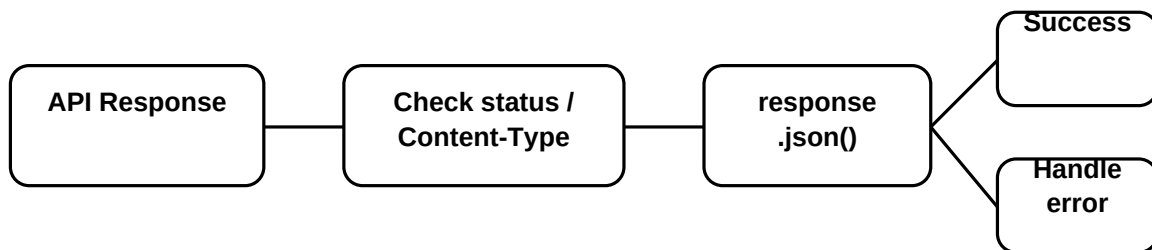


Example: If the API says Retry-After: 10, wait about 10 seconds and then retry the request.

Key Point: Handle API errors, invalid responses, and missing data safely.

2. If `response.json()` fails, what should your program do?

Answer: The program should handle the JSON parsing failure safely instead of crashing. An API may return HTML, plain text, an empty response, or another invalid format. The program should check the response status and, where appropriate, the **Content-Type**. JSON parsing should be inside exception handling. If parsing fails, log the problem and show a useful fallback or error message.



Handle invalid or non-JSON responses safely

Example: A server error page may be HTML, so `response.json()` can fail. The program should handle that case gracefully.

Key Point: Handle API errors, invalid responses, and missing data safely.

3. If a JSON key is missing, why might `.get()` be safer than `[]`?

Answer: In Python, `data.get('key')` is safer when a JSON field may be missing because it returns **None** by default, or a value supplied as a default. In contrast, `data['key']` raises a **KeyError** when the key does not exist. Therefore, `.get()` helps the program continue safely when optional data is unavailable.

JSON data

`data.get('name')`
Missing → None / default

SAFER for optional fields

`data['name']`
Missing → **KeyError**

ERROR if key is missing

Example: `name = data.get('name', 'Unknown')` safely gives 'Unknown' when the name field is absent.

Key Point: Handle API errors, invalid responses, and missing data safely.